

Other financial models may give us additional insight into the workings of modern finance. Option-pricing theory began in 1900 when Bachelier saw how options could control risk and tried to figure out how to price them. Bachelier reasoned that if an option was a “fair bet,” then it had to have a fair value. He was not entirely successful in computing fair option values, but he was not far off either. Subsequent efforts to price options were complex or awkward, and they often depended on individual risk preferences.

Since the time of Irving Fisher, economists have been enamored with the tidiness of equilibrium-based models. Inspired by Bachelier’s random walk ideas, Thorp and Kassouf (1967) came up with an equilibrium-based options model, but their published work did not discount back the expected value of an option at expiration.

It was Fischer Black and Myron Scholes who published the first complete equilibrium-based option-pricing model. It resulted in Nobel prizes, since the economic community loves equilibrium-based models. The Black-Scholes option-pricing model (the BS model) became a favorite and a showpiece among financial economists.

Option-pricing models can be useful in helping users transfer risk by using derivatives. Investors were quick to recognize this. In 1970, there was almost no trading in financial derivatives. By 2004, outstanding derivative contracts totaled \$273 trillion.

Besides increasing the popularity of derivatives, pricing models also gave some users a false sense of security. As we will see in [Chapter 4](#), we tend to overweight our own skills and knowledge, which causes us to suffer from an illusion of control and underestimate the odds of bad events and unfavorable outcomes.

In addition to the near meltdown of the world’s financial system in 1998 due to the Long-Term Capital Management (LTCM) debacle, derivatives contributed to the bankruptcy of Orange County, California, the collapse of Barings Bank, and, of course, the global financial crisis of 2007–2008. Warren Buffett referred to derivatives as “financial weapons of mass destruction.”

Institutional investors were slow to learn their lessons. Following the collapse of LTCM in 1998, Merrill Lynch warned that mathematical risk models “may provide a greater sense of security than warranted; therefore,